

EXPERIMENT – 9

ULTRASONIC SENSOR IN PROTEUS USING ARDUINO

AIM: To write an Embedded C program for interfacing Ultrasonic Sensor using Arduino Uno and Proteus.

OBJECTIVES:

1. To interface the ultrasonic sensor with Arduino Uno.
2. To measure distance using trigger and echo signals.
3. To display the measured distance through serial communication.

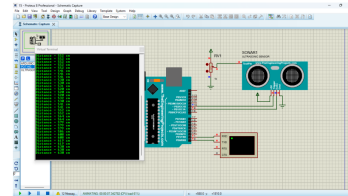
PROGRAM:

```
int trig = 10;
int echo = 9;
long duration;
int cm;
void setup()
{
    pinMode(trig, OUTPUT);
    pinMode(echo, INPUT);
    Serial.begin(9600);
}
void loop()
{
    digitalWrite(trig, LOW);
    delayMicroseconds(10);
    digitalWrite(trig, HIGH);
    delayMicroseconds(10);
    digitalWrite(trig, LOW);
    delayMicroseconds(10);
    duration = pulseIn(echo, HIGH);
    cm = (duration / 2) * 0.034;
    Serial.print(" Distance = ");
    Serial.print(cm);
    Serial.println(" cm");
}
```

OUTPUT IMAGE:

```
Serial.begin(9600);  
}  
  
void loop() { // Sketch to calculate ultrasonic distance  
  digitalWrite(trig, LOW);  
  delayMicroseconds(10);  
  digitalWrite(trig, HIGH);  
  delayMicroseconds(10);  
  digitalWrite(trig, LOW);  
  delayMicroseconds(10);  
  duration = pulseIn(echo, HIGH);  
  cm = (duration/2) * 0.034;  
  // Print Value on Serial Monitor  
  Serial.print(" Distance = ");  
  Serial.print(cm);  
  Serial.println(" cm");  
}
```

OUTPUT:



RESULT:

Thus the program has been successfully verified and executed.

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